AHMED IQBAL

AI Engineer | NLP & LLM Engineer | Fine-Tuning

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PROFESSIONAL SUMMARY

AI Engineer specialising in Natural Language Processing and Large Language Model systems, with hands-on work across the full NLP pipeline: collecting and scraping data from heterogeneous web and document sources, cleaning and preprocessing text, preparing and validating labelled datasets, fine-tuning Transformer models with Hugging Face and PyTorch, and evaluating them against held-out data. Practical experience in text classification, information extraction, PDF/DOCX document intelligence, embeddings, semantic search and retrieval-augmented generation for grounded, source-cited answers. Builds and maintains production AI services in Python with FastAPI and Flask, and independently researches and implements current language-modelling techniques to improve domain-specific model quality.

TECHNICAL SKILLS

NLP & LLM Natural Language Processing, Large Language Models, Transformer Architecture, Hugging Face Transformers, Fine-Tuning, Text Classification, Information Extraction, Named Entity Recognition, Embeddings, Semantic Search, Prompt Engineering

ML / Deep Learning PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Model Training, Model Evaluation, Deep Neural Networks, CNNs, Vision Transformers

LLM Systems Retrieval-Augmented Generation (RAG), Vector Retrieval, LangChain, LangGraph, LLM Evaluation, Document Intelligence, Agentic Workflows

Data & Documents Web Scraping, Data Extraction, PDF/DOCX Parsing, Text Extraction, Data Cleaning, Data Preprocessing, Dataset Preparation, Labelling & Validation

Backend Python, FastAPI, Flask, REST APIs, JavaScript, SQL

Databases & Retrieval FAISS, ChromaDB, PostgreSQL, MySQL, SQLite, MongoDB

Development Git, Docker, Jupyter,

PROFESSIONAL EXPERIENCE

AI Engineer — Apr 2026 – Present

DX Creativ Digital Marketing & Agency · On-site

- Develop and optimise LLM-driven agent systems that automate web-development workflows, iterating on prompt design, tool orchestration and output-quality evaluation to raise task success rates, built and deployed two client WordPress websites, QuickJump-Start and GlanPlawPets, through the AI agent.
- Maintain and extend an existing production AI codebase, refactoring pipeline components and resolving failure modes surfaced during real client usage and websites.
- Analyse campaign and search datasets to diagnose weak model outputs for SEO and Google Ads agents, feeding the findings back into prompt, retrieval and configuration improvements.

AI Full Stack Engineer – Intern — Dec 2025 – Feb 2026

Arfa Kareem Technologies · Peshawar, Pakistan · On-site

- Built production-oriented NLP and machine-learning applications in Python (Flask, FastAPI) for international and local clients, owning each system end-to-end from raw data collection through preprocessing, model training, evaluation and deployment.
- Engineered a document-grounded Retrieval-Augmented Generation system over enterprise PDF and document parsing, chunking, embedding generation, vector indexing and top-k retrieval so that LLM answers were constrained to retrieved source passages instead of parametric memory.
- Collected and normalised training and evaluation data from heterogeneous web pages and documents using Python scraping and parsing, resolving inconsistent layouts, encodings and duplicate records before dataset construction.
- Developed data preprocessing and feature pipelines for an AI Running Coach Assistant, training and comparing multiple supervised models with Scikit-learn and tuning them against held-out validation data to improve prediction accuracy.
- Implemented multi-agent LLM workflows with LangGraph for structured multi-step information processing, including explicit state handling and controlled tool invocation for reliable, inspectable execution.

Technologies: Python, Flask, FastAPI, LangChain, LangGraph, RAG, LLMs, Scikit-learn, Pandas, Vector Databases, REST APIs, Git

Junior Data Scientist – Intern — Dec 2024 – Feb 2025

Cognifyz Technologies · Remote

- Performed data cleaning, preprocessing and exploratory data analysis on real-world datasets, handling missing values, outliers and inconsistent schemas prior to modelling.
- Built and evaluated supervised machine-learning models with Scikit-learn, improving classification accuracy from 87% to 92% through feature engineering and hyperparameter tuning.

RELEVANT AI / NLP PROJECTS

AcademicGuard — AI Research Integrity Platform **NLP · Document Intelligence**

- Building an NLP system that classifies AI-generated versus human-written text in research papers, preparing and balancing labelled text datasets and implementing tokenisation, normalisation and cleaning pipelines for Transformer input.
- Fine-tuned Transformer-based NLP models using Hugging Face and PyTorch for text classification and hands-on experience with Transformer-based model training, Hugging Face, PyTorch and evaluated checkpoints on held-out splits using accuracy, precision, recall and confusion-matrix analysis.
- Implemented document parsing and text extraction for submitted PDF and DOCX papers, recovering clean, section-aware text for downstream classification and summarisation.
- Integrated LLM-based summarisation and structured report generation to turn raw model outputs into reviewer-facing integrity reports with per-section findings.

Technologies: Python, PyTorch, Hugging Face Transformers, Vision Transformer, TensorFlow, Scikit-learn, CNNs, Flask, React, SQLite

YouTube Knowledge Assistant — Knowledge-Grounded Retrieval System **RAG · Semantic Search**

- Built an ingestion pipeline that transcribes lecture video, segments and cleans the resulting transcripts, and generates embeddings for semantic retrieval over the video corpus.
- Implemented vector search with ChromaDB and deliberate context construction so that generation is grounded strictly in retrieved transcript passages, suppressing hallucinated answers outside the source material.
- Evaluated retrieval relevance and answer faithfulness against the source transcripts and served the system through a FastAPI backend; reduced lecture revision time by 50% in practical use.

Technologies: Python, LangChain, ChromaDB, Embeddings, Semantic Search, FastAPI, Streamlit/Flask · Demo

Pharma Clinical Decision Support System **LLM Application · Desktop**

- Developed an AI-powered clinical decision support system that analyses patient records, laboratory results and prescribed medications to generate drug-interaction, side-effect and risk-assessment reports for pharmacists.
- Built the backing web platform for administration, user monitoring and analytics, with a Python/FastAPI service layer over PostgreSQL.

Technologies: Python, FastAPI, Flask, LangChain, PostgreSQL, React, Electron.js, REST APIs

Autonomous Academic Workflow Agent **Agentic LLM Workflow**

- Built an autonomous LLM agent that monitors Google Classroom assignments in real time, generates candidate solutions and routes them through generation, review, regeneration and submission stages.
- Designed human-in-the-loop approval checkpoints so a user can approve, reject or regenerate any model output before it is submitted.

Technologies: Python, LangGraph, LangChain, FastAPI, Google Classroom API, Google Drive API

EDUCATION

BS, Software Engineering — CGPA 3.94 / 4.00 (Gold Medalist) **2022 – 2026**

CECOS University of IT & Emerging Sciences, Peshawar, Pakistan

- Relevant coursework: Artificial Intelligence, Natural Language Processing, Algorithms, Database Systems, Operating Systems, Computer Networks, Calculus, Statistics & Probability.

CERTIFICATIONS

Machine Learning — Stanford University (Coursera) | Google Advanced Data Analytics

IBM Data Science Professional Certificate | IBM Data Analyst Professional Certificate